(12) UK Patent Application (19) GB (11) 2 138 123 A

(43) Application published 17 Oct 1984

(21) Application No 8408973

(22) Date of filing 6 Apr 1984

(30) Priority data

(31) 3313888 (32) 16 Apr 1983 (33) DE

(51) INT CL³
F16T 1/08

(52) Domestic classification
F4U 31
F2V S23

(56) Documents cited
GB A 2117108 GB A 2067741 GB A 2064723
GB A 2054828

(58) Field of search
F4U

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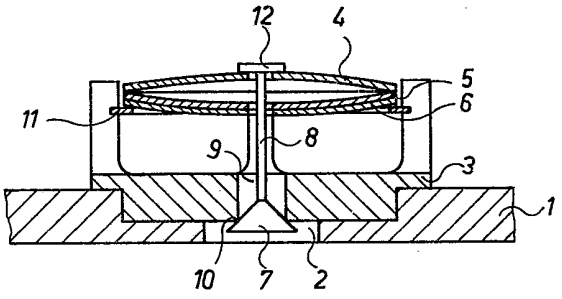
(54) Steam trap

(57) A steam trap has a closure part (7) arranged on the low pressure side, which co-operates with a valve seat (10) and which is operated, through a shaft (5), by a control unit comprising bimetallic snapping members (4, 5, 6) arranged in the high pressure space of the steam trap.

The bimetallic snapping members (4, 5, 6) are engaged directly or by means of webs at one end on fixed abutments (11), and at the other end on an abutment (12), fitted to the shaft (8). The abutments (11, 12) are arranged in such a way that the closure member (7) engages on the valve seat (10) within the snapping stroke of the individual bimetallic snapping members (4, 5, 6).

The steam trap can be used in a wide pressure range, without inadmissibly high stresses occurring in the bimetallic snapping members (4, 5, 6).

Fig. 2



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Fig. 1

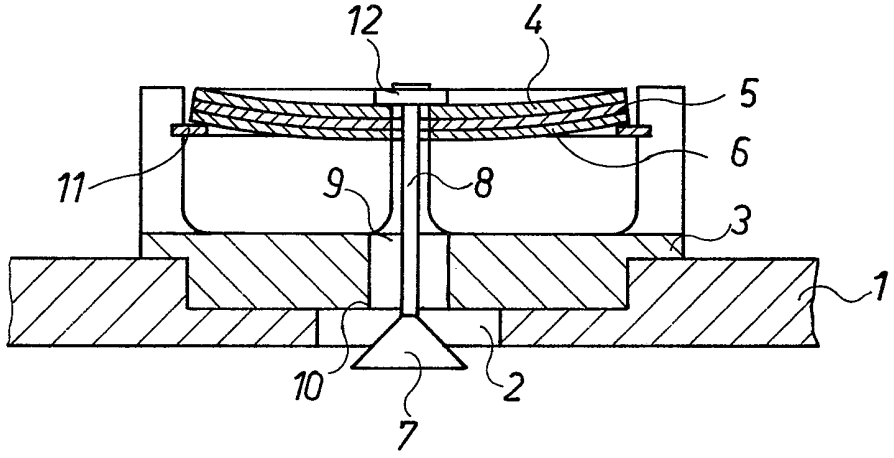


Fig. 2

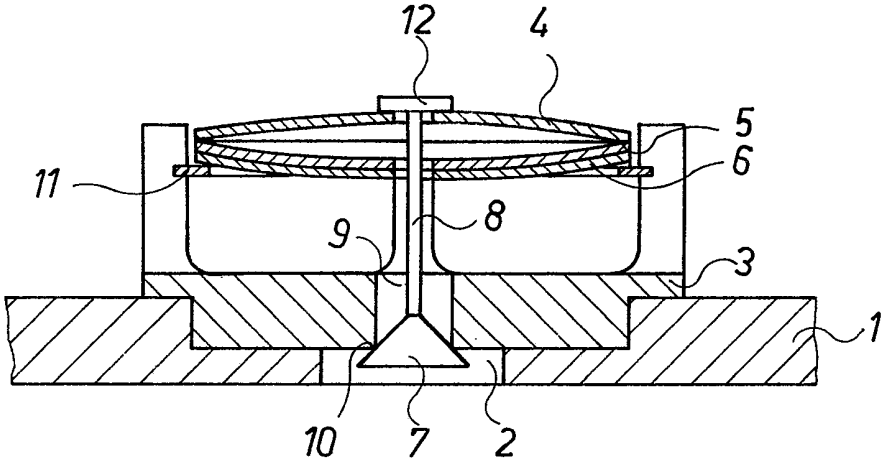


Fig. 3

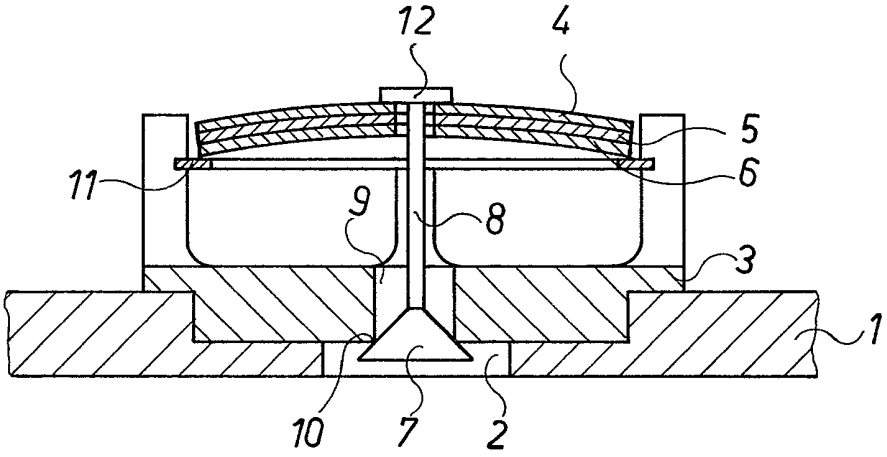


Fig. 4

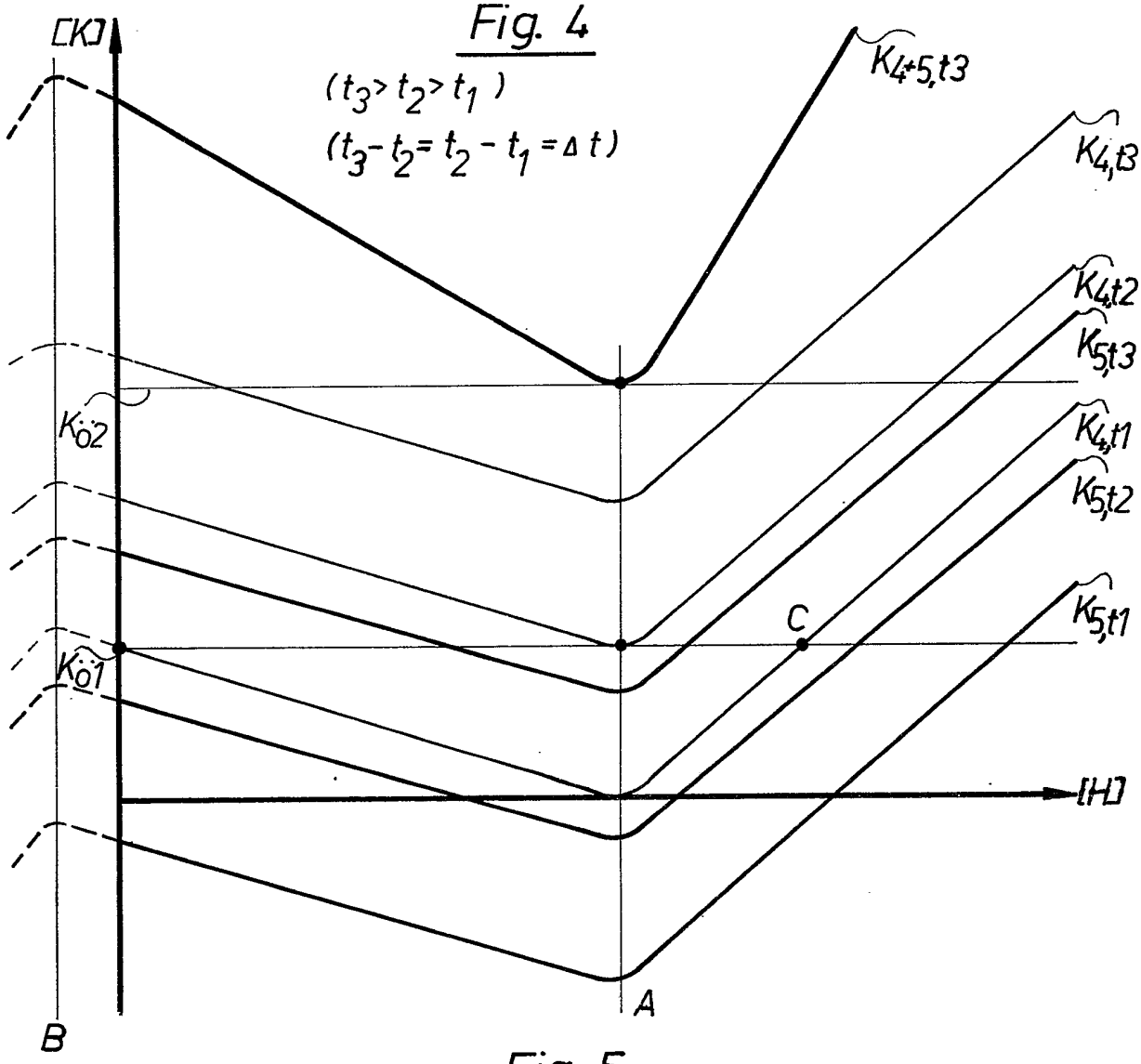
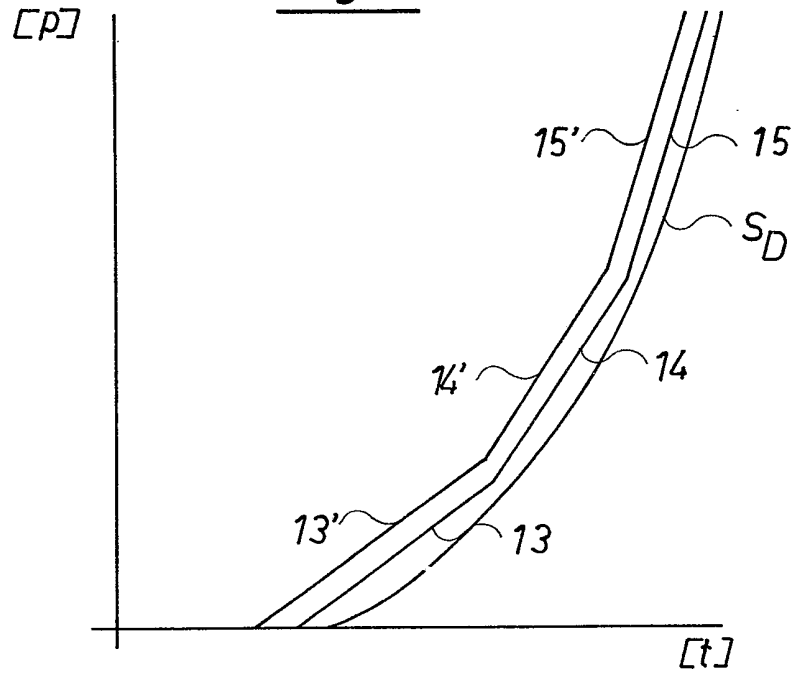

[K]
$(t_3 > t_2 > t_1)$
$(t_3 - t_2 = t_2 - t_1 = \Delta t)$
$K_{4+5,t3}$
$K_{4,t3}$
$K_{4,t2}$
$K_{5,t3}$
$K_{4,t1}$
$K_{5,t2}$
$K_{5,t1}$
$K_{\ddot{o}2}$
$K_{\ddot{o}1}$
C
[H]
A
B

Fig. 5


[p]
15'
15
S_D
14'
14
13'
13
[t]

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Fig.6

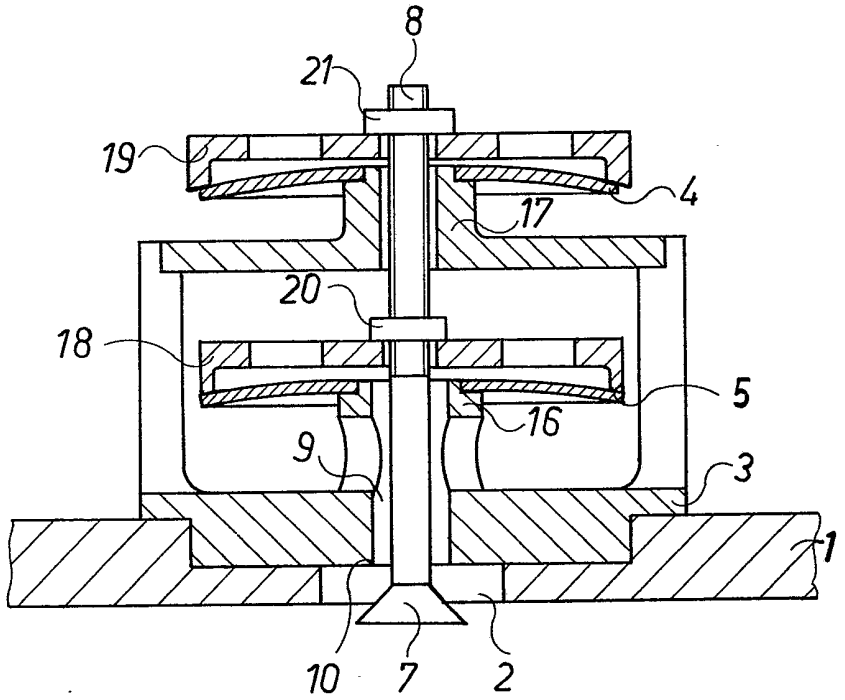
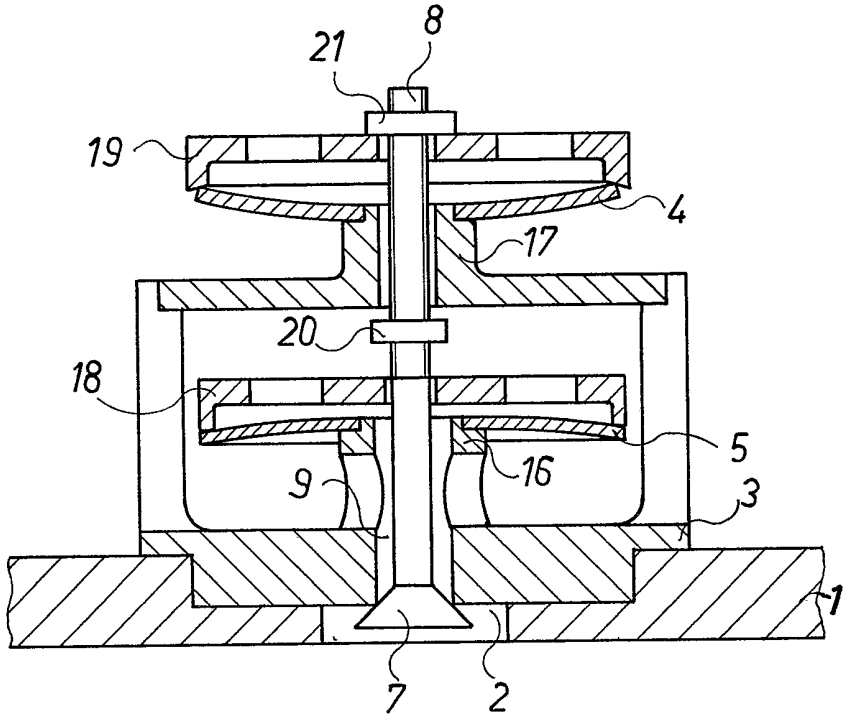


Fig.7



SPECIFICATION

Steam trap

This invention relates to steam traps and is particularly concerned with a steam trap having a closure part arranged to co-operate with a valve seat and operated by a control unit.

In order to prevent erosion of the closure part and to reduce the susceptibility to dirt, every effort is made in connection with steam traps to open them as widely as possible after initiating the opening process and then to close them completely again after draining off the condensed steam. It is desirable to maintain the function within a wide pressure range, so that only a minimum number of types has to be kept in stock.

A steam trap is known from German Offenlegungsschrift No. 29 48 686, in which use is made of bimetallic snapping members. On reaching predetermined temperatures in the case of this steam trap, the closure part snaps into the closed position and then snaps out of this position again after a certain degree of cooling. The bimetallic snapping members are arranged in such a way that their deflections are summated, all of them being exposed to the same force flow. The weaker members perform the snapping process at lower temperatures and pressures, whereas the stronger members are used for higher pressures. The latter then simultaneously prevent any snapping process of the weaker members. As a result, the weaker bimetallic snapping members are exposed to extremely high stresses at high temperatures and pressures, so that the pressure range to which this principle can be applied remains limited.

The present invention aims to provide a steam trap of the aforementioned type, which can be used in a wide pressure range, without the bimetallic snapping members being exposed to inadmissibly high stresses.

According to the invention, there is provided a steam trap having a closure part arranged on the low pressure side, which is arranged to co-operate with a valve seat and to be operated by a control unit, wherein the control unit comprises a plurality of bimetallic snapping members arranged in the high pressure space of the steam trap by means of a shaft, the bimetallic snapping members being supported directly or by means of webs with one end on fixed abutments and the other end on abutments fitted to the shaft, the ends of the bimetallic snapping members associated with the shaft being arranged to move in the closing direction in the case of a temperature rise and the abutments being arranged in such a way that the closure part engages on the valve seat within the snapping stroke of the individual bimetallic snapping members.

The forces of the individual bimetallic snapping members act in a summated manner on the closure part, as soon as their snapping temperatures are exceeded.

However, as the bimetallic snapping members are independent of one another with respect to the force flow from the fixed abutment thereof to the abutment on the shaft or shank of the closure part, the weaker bimetallic snapping members are not subject to the action of the high forces of the stronger bimetallic snapping members. Thus, the steam trap can be used in a wide pressure range, without there being inadmissibly high stresses in the bimetallic snapping members.

According to a preferred embodiment of the invention, the minima of the spring tension isotherms of the individual bimetallic snapping members are at the same stroke position and the maxima of the spring tension isotherms of the individual bimetallic snapping members are at the same stroke position, which differs from the stroke position at said minima, with respect to the closing position of the closure part. Since the spring tension isotherms are made to coincide in this embodiment, the difference between maximum and minimum responsible for the snapping process is most marked at tension summation.

The position of the maxima of the spring tension isotherms may correspond to or be beyond the closed position of the closure part. These features not only lead to the maximum sealing forces on the closure part, on closing, but in the case of a temperature drop bring about a sudden opening, without any prior slow-speed movement.

Preferably, the bimetallic snapping members have different snapping temperatures. As a result of this, the bridging of a large operating temperature range is not brought about by a linear force or tension increase, but instead there is a polygonal configuration of associated tensions and snapping temperatures, so that the steam trap opens and closes with an almost constant temperature spacing of the steam tension curve.

The different bimetallic snapping members may have different specific thermal deflections. This makes it possible to keep the shape and/or geometry of the different bimetallic snapping members as identical as possible, all within a favourable constructional scope.

A plurality of fixed abutments and a plurality of abutments on the shaft may be associated with the bimetallic snapping members, the abutments being adjustable. This arrangement makes it possible to adjust individually the bimetallic snapping members via the abutments associated therewith and permits less stringent requirements regarding the manufacturing precision of the bimetallic snapping members, because e.g. snapping stroke ranges differing from the design ranges can be compensated by relative adjustment of the abutments.

Finally, according to one embodiment of the invention, each of the bimetallic snapping members of the control unit is fixed at one end thereof while the other end acts either directly or via a respective compressively rigid web on an abutment on the shaft in such a way that in a first part of the complete working range of the steam trap at least one bimetallic snapping member with a lower snapping temperature exerts a closing force on the closure part while in other parts of the working range of the steam trap, in addition to the member with the lower snapping temperature, at least one bimetallic snapping member with a higher snapping temperature exerts an additional closing force on the closure part.

The invention will now be further described by way of example, with reference to the drawings, in

which:—

Fig. 1 is a vertical section through a first embodiment of a steam trap according to the invention having a control unit shown in the open position;

Fig. 2 is a view, corresponding to Fig. 1, but showing the control unit in the closed position and with a low operating pressure;

Fig. 3 is a view, corresponding to Fig. 1, but showing the control unit in the closed position and with a high operating pressure;

Fig. 4 is a graph of the bimetallic spring tension isotherms, plotted over the closure part stroke;

Fig. 5 is a diagram of a saturated steam curve, as well as the closing and opening temperatures;

Fig. 6 is a vertical section through a second embodiment of a steam trap according to the invention having a control unit shown in the open position; and

Fig. 7 is a view, corresponding to Fig. 6, but showing the control unit in the closed position and with a low operating pressure.

Referring to the drawings, a partition 1 between the inlet pressure and low pressure side of a steam trap casing (not shown) has a passage bore 2 for the condensed steam to be drained off and carries a control unit. The latter comprises a seating member 3, three disc-shaped, centrally perforated bimetallic snapping members 4, 5, 6 arranged on the inlet pressure side and a closure part 7 with a shank or shaft 8 arranged on the low pressure side. The seating member 3 has a central passage 9 and a valve seat 10, with which co-operates the closure part 7.

Figs. 1 to 3 relate to a seating member 3 with a fixed abutment 11, on which rests the outer edge of the bimetallic snapping members 4, 5, 6, whilst their inner edge engages on an abutment 12 provided on the shank or shaft 8.

In the cold state, all three bimetallic snapping members 4, 5, 6 are convexly curved towards the valve seat (Fig. 1). The closure part 7 opens the passage 9 and condensed steam can flow out. With rising temperature, the bimetallic snapping members 4, 5, 6 initially move slowly in the closing direction until their particular snapping temperatures are reached and then spring through.

In the case of bimetallic snapping members 4, 5, 6 with different snapping temperatures, the opening and closing processes are shown in the graph shown in Fig. 4. The strokes H of the bimetallic snapping members 4, 5, 6 as well as of the closure member 7 are plotted on the abscissa. The co-ordinate zero corresponds to the closed position. The forces or tensions are plotted on the ordinate, K4 indicating the spring tensions which can be exerted by the bimetallic snapping member 4 on the closure member 7 at the exemplified temperatures t1 to t3, whilst K5 indicates the corresponding spring tensions of the bimetallic snapping member 5. The curves plotted in the graph for K4 and K5 form the spring tension isotherms. The spring tensions K4, K5 act in the closing direction for as long as the spring tension isotherms are on the positive ordinate. They theoretically act in the opening direction when they are negative. As the bimetallic snapping members 4, 5, 6 only engage on one side on the abutments 11, 12 said negative forces cannot be transferred to the closure part 7.

Kö is the force determined by the pressure gradient on the closure part 7 and acting on the latter in the opening direction, it being essentially independent of the stroke or travel.

The position or stroke of the closure part 7 is determined by the equilibrium of the opening pressure force Kö and the spring tensions K4, K5, K6 of the bimetallic snapping members 4, 5, 6 or the sums thereof and can only be at the intersection point of the force lines for the prevailing temperatures and pressures.

Thus, if in the case of opening forces up to Kö1 the temperature rises from t1 to t2, the bimetallic snapping member 4 inches the closure part 7 from the open position (to the right in the graph) to the stroke position A and from the the latter suddenly brings it into the closed position. However, the bimetallic snapping members 5, 6 remain in a form convexly curved towards the valve seat 10 (Fig. 2). The forces K4 associated with the different closing snapping temperatures are represented on the perpendicular A passing through the minima of the spring tension isotherms. In the graph of Fig. 5, this association leads to a first line 13.

In the case of temperatures higher than t2, e.g. at t3, the bimetallic snapping member 5 is also in a position to apply positive forces, i.e. acting in the closing direction over the entire stroke range. The spring tension isotherms of the bimetallic snapping members 4 and 5 are consequently summated to the resultant "K4 + 5, t3". If the temperature rises to t3 at higher opening pressure forces than Kö1, e.g. at Kö2, the bimetallic snapping members 4, 5 jointly inch the closure part 7 from the open position to position A and then snap in the closed position. The bimetallic snapping member 6 remains in its convexly curved form with respect to valve seat 10.

Thus, the forces of the bimetallic snapping members 4 and 5 are summated on the perpendicular through A, so that in the case of the same t, the force difference increases. In Fig. 5 there is a second line 14 with a sharper gradient. This is repeated when the bimetallic snapping member 6 also comes into action (Fig. 3), so that there is a third, even steeper line 15 of the traverse shown in Fig. 5.

It can be gathered from Fig. 4 that the opening process represents a reversal of the closing process. If the maxima of the spring tension isotherms correspond to the closing position or are beyond the latter, e.g. on the perpendicular through B, the steam trap suddenly opens from the closed position to point C, e.g. at pressure force Kö1, as soon as the temperature of the bimetallic snapping member 4 has dropped from t2 to t1. Thus, Fig. 5 contains a second traverse 13', 14', 15' for the opening process, which merely has a somewhat lower temperature. Curve SD in Fig. 5 gives the saturated steam temperature representing the latest time at which the steam trap can be closed, in the case of different pressures.

The graph of Fig. 5 shows that the sought after action is achieved over a wide pressure range. This is possible because the individual bimetallic snapping members 4, 5, 6 are independent of one another with respect to the force flow from the fixed abutment 11 to

the abutment 12 on the closure part shank or shaft 8. Thus, the weaker bimetallic snapping members 4, 5 are not under the action of the high spring tensions of the stronger bimetallic snapping members 5, 6 and are therefore not exposed to inadmissibly high stresses.

In the embodiment according to Figs. 6 and 7 the seating member 3 carries two central, fixed abutments 16, 17. The inner edge of a bimetallic snapping member 4, 5 is supported on each of them. Dish-shaped webs 18, 19 engage on the outer edges of the two bimetallic snapping members 4, 5 and their inner edge acts on two abutments 20, 21 arranged on the shank or shaft 8 of the closure part. These are made axially displaceable for adjusting the steam trap.

As described in connection with Figs. 1 to 3, in the case of a rising temperature, the bimetallic snapping members 4, 5 initially move slowly in the closing direction from the open position. On reaching its associated snapping temperature, the bimetallic snapping member 4 snaps through and by means of the web 19 and abutment 21, raises the closure part 7 on its shaft 8 into the closed position. The bimetallic snapping member 5 remains in its form concavely curved towards valve seat 10, so that during the closing stroke of the closure part 7, the abutment 20 rises from the web 18 (Fig. 7). In the case of higher opening forces of the closure part 7, i.e. at higher pressures and correspondingly higher temperatures, both the bimetallic snapping members 4 and 5 snap through and respectively engage the webs 18, 19 with abutments 20, 21, so that both their closing forces act in a summated manner on the closure part 7. What was stated hereinbefore regarding the adaptation of the traverse of snapping lines 13, 14 of the bimetallic snapping members 4, 5 to the steam tension curve SD then applies.

CLAIMS

1. A steam trap having a closure part arranged on the low pressure side, which is arranged to co-operate with a valve seat and to be operated by a control unit, wherein the control unit comprises a plurality of bimetallic snapping members arranged in the high pressure space of the steam trap by means of a shaft, the bimetallic snapping members being supported directly or by means of webs with one end on fixed abutments and with the other end on abutments fitted to the shaft, the ends of the bimetallic snapping members associated with the shaft being arranged to move in the closing direction in the case of a temperature rise and the abutments being arranged in such a way that the closure part engages on the valve seat within the snapping stroke of the individual bimetallic snapping members.

2. A steam trap according to claim 1, wherein the minima of the spring tension isotherms of the individual bimetallic snapping members are at the same stroke position and the maxima of the spring tension isotherms of the individual bimetallic snapping members are at the same stroke position, which differs from the stroke position at said minima, with respect to the closing position of the closure part.

3. A steam trap according to claim 2, wherein the position of the maxima of the spring tension isotherms corresponds to the closed position of the closure part.

4. A steam trap according to claim 2, wherein the maxima of the spring tension isotherms are beyond the closed position of the closure part.

5. A steam trap according to any preceding claim, wherein the bimetallic snapping members have different snapping temperatures.

6. A steam trap according to claim 5, wherein the different bimetallic snapping members have different specific thermal deflections.

7. A steam trap according to any preceding claim, wherein a plurality of fixed abutments and a plurality of abutments on the shaft are associated with the bimetallic snapping members, the abutments being adjustable.

8. A steam trap according to any preceding claim, wherein each of the bimetallic snapping members of the control unit is fixed at one end thereof while the other end acts either directly or via a respective compressively rigid web on an abutment on the shaft in such a way that in a first part of the complete working range of the steam trap at least one bimetallic snapping member with a lower snapping temperature exerts a closing force on the closure part while in other parts of the working range of the steam trap, in addition to the member with the lower snapping temperature, at least one bimetallic snapping member with a higher snapping temperature exerts an additional closing force on the closure part.

9. A steam trap substantially as described herein with reference to the drawings.

Printed in the United Kingdom for Her Majesty's Stationery Office, 8818935, 10/84, 18996. Published at the Patent Office, 25 Southampton Buildings, London WC2A 1AY, from which copies may be obtained.